



Chitkara

PAPER OF CSE-2025 - PYTHON TEST1 EXAM

Q1. Python Coding

In the kingdom of **Numberland**, two warriors — each represented by a number — enter the **Arena of Comparison**. The one with the **greater strength (value)** becomes the Champion.

As the royal programmer, your task is to write a Python program that helps the king determine which warrior is stronger. You will be given two integers on a single line (space-separated), representing the strength of the two warriors. Your program should output the **larger number**.

If both warriors have equal strength, they are considered equals, and you can print either of the values (since they're the same).

No extra messages or text should be printed — only the number representing the stronger warrior.

Input Format:

Two integers in different lines.

Output Format:

A single integer – the larger of the two numbers.

Sample Input

```
42
77
```

Sample Output

```
77
```

ANSWER:

Test case 1 :

```
Input:
42
77
Output:
77
```

Test case 2 :

```
Input:
45
98
Output:
98
```

Test case 3 :

```
Input:
13
9
Output:
13
```

Test case 4 :

```
Input:
99
100
Output:
100
```

Test case 5 :

```
Input:
5
5
Output:
5
```

Test case 6 :

```
Input:
44
5
Output:
44
```

Q2. Python Coding

One sunny afternoon, a young student Manu sat at their computer, wondering if they were old enough to vote. They decided to ask their trusty program for the answer. "What is your

age?" the program asked. The student entered their age, hoping for an answer. The program thought for a moment and said, "If your age is 18 or older, you are eligible to vote." The student waited nervously. If the number entered was 18 or more, the program would announce, "Eligible to vote!" But if it was less, it would respond with a simple, "Not eligible to vote."

INPUT:

The program accepts a single integer input from the user, representing their age.

OUTPUT:

The program outputs one of two possible messages:
Eligible to vote. // if the age is greater than or equal to 18.
Not eligible to vote. // if the age is less than 18.

SAMPLE INPUT:

18

SAMPLE OUTPUT:

Eligible to vote.

ANSWER:Test case 1 :

Input:

18

Output:

Eligible to vote.

Test case 2 :

Input:

17

Output:

Not eligible to vote.

Test case 3 :

Input:

25

Output:

Eligible to vote.

Test case 4 :

Input:

16

Output:

Not eligible to vote.

Test case 5 :

Input:

40

Output:

Eligible to vote.

Test case 6 :

Input:

0

Output:

Not eligible to vote.

Q3. Python Coding

A school is planning to buy notebooks for students from one of two suppliers.

Each supplier has quoted a price for a single notebook:

- Supplier A charges **X** rupees per notebook
- Supplier B charges **Y** rupees per notebook

To reduce expenses, the school must choose the supplier offering the **lower price**.

Given the quoted prices **X** and **Y**, determine which supplier is more economical by reporting the **lowest price among the two**.

Input

Each test case contains 2 numbers a and b, separated by a space where a not equal to b.

Output

Print the smallest number.

Sample Input 1

10

20

Sample Output 1

10

ANSWER:Test case 1 :

Input:

10

20

Output:

10

Test case 2 :

Input:

80

70

Output:

70

Test case 3 :

Input:

33

34

Output:

33

Test case 4 :

Input:

20

19

Output:

19

Test case 5 :

Input:

45

66

Output:

45

Test case 6 :

Input:

88

33

Output:

33

Q4. Python Coding

Riya is playing with two numbers.

She wants to do the following:

If the **product** of the two numbers is **even**, print **the sum** of the numbers.

If the **product** of the two numbers is **odd**, print **the difference** (first number – second number).

Sample Input 1

4

5

Sample Output 1

9

Explanation:

$4 \times 5 = 20$ (even), so we print $4 + 5 = 9$.

Sample Input 2

7

3

Sample Output 2

4

Explanation:

$7 \times 3 = 21$ (odd), so we print $7 - 3 = 4$.

ANSWER:

Test case 1 :

Input:

4

5

Output:

9

Test case 2 :

Input:

7

3

Output:

4

Test case 3 :

Input:

9

10

Output:

19

Test case 4 :

Input:

15

951

Output:

-936

Test case 5 :

Input:

489

154489

Output:

-154000

Test case 6 :

Input:

49

10

Output:

59

Test case 7 :

Input:

4800

2201

Output:

7001

Q5. Python Coding

A major bank is flooded with thousands of loan applications every day. Employees are struggling with the paperwork, and frustrated customers are lining up in long queues. To fix this chaos, the bank decides to automate the loan approval process. But the applicants don't always cooperate:

- Some type their age as negative numbers like -5, or even unrealistic values like 200.
- Some type their age or credit score as words like "twenty-five" instead of numbers.
- Others submit a credit score of 0 or leave it blank, hoping the system won't notice.

The bank sets clear rules for automation:

- Applicants must be at least 21 years old.
- Applicants must have a credit score greater than 700.
- If invalid input (negative, zero, or text) is entered → the system should output "Invalid input".

Your job is to build the program that will help the bank decide whether the loan is approved, rejected, or invalid. Decision Rules

- If $\text{age} < 21 \rightarrow \text{"Loan Rejected"}$
- If $\text{credit score} \leq 700 \rightarrow \text{"Loan Rejected"}$
- If invalid input (negative, zero, or non-numeric) $\rightarrow \text{"Invalid input"}$
- If both conditions are satisfied ($\text{age} \geq 21$ and $\text{credit} > 700$) $\rightarrow \text{"Loan Approved"}$

Input Format

First line: Age (integer)

Second line: Credit score (integer)

Output Format

A decision string

Sample Input

25
720

Sample Output

Loan Approved

ANSWER:

Test case 1 :

Input:

25

720

Output:

Loan Approved

Test case 2 :

Input:

18

750

Output:

Loan Rejected

Test case 3 :

Input:

30

650

Output:

Loan Rejected

Test case 4 :

Input:

-5

800

Output:

Invalid input

Test case 5 :

Input:

twenty

720

Output:

Invalid input

Test case 6 :

Input:

30

900

Output:

Loan Approved

Q6. Python Coding

A student was given a task to automate the system that takes a number (n) as an input, finds the first and last digit of that number and calculates the multiplication of these two digits. The calculated product is produced as an output by the system.

Write a code to automate this system.

Sample Input 1

9379

Sample Output1

81

Sample Input 2

8345

SampleOutput2

40

Input Explanation:

Input any positive number

Output Explanation:

Prints the product of first and last digit of a number

ANSWER:

Test case 1 :

Input:

9379
Output:
81

Test case 2 :

Input:
8345
Output:
40

Test case 3 :

Input:
9375
Output:
45

Test case 4 :

Input:
2500
Output:
0

Test case 5 :

Input:
8345
Output:
40

Test case 6 :

Input:
6703
Output:
18

Test case 7 :

Input:
1234
Output:
4

Q7. Python Coding

A mathematician has a requirement to design a system where the system takes an input a number of n digits and finds out the largest digit from that number. The largest digit is then displayed as an output by the system.

Write a code to develop this system to fulfill his requirements.

Sample Input 1

```
46749 //Number
```

Sample Output1

```
9 //Largest digit in the number
```

Sample Input 2

```
386748
```

SampleOutput2

```
8
```

Input Explanation:

Input any number

Output Explanation:

Displays the largest digit in a given number of n digits.

ANSWER:

Test case 1 :

Input:
46749
Output:
9

Test case 2 :

Input:
386748
Output:
8

Test case 3 :

Input:
46759
Output:
9

Test case 4 :

Input:
387748
Output:
8

Test case 5 :

Input:
24706
Output:
7

Test case 6 :

Input:
244502
Output:
5

Test case 7 :

Input:
10003
Output:
3

Q8. Python Coding

The national tax office is digitizing millions of account numbers. To quickly validate them, they want a system that sums up all digits of an account number.

But, as always, disruptions creep in:

- Some taxpayers accidentally enter negative account numbers.
- Some enter very long numbers beyond the standard length.
- Some even type extra characters or symbols that don't belong.
- The system must ignore negative signs and focus only on digits.

Your task is to build a program that extracts digits and returns their sum.

Input Format

A single integer (positive or negative).

Output Format

An integer → the sum of digits.

Sample Input

1234

Sample Output

10

ANSWER:

Test case 1 :

Input:
1234
Output:
10

Test case 2 :

Input:
-567
Output:
18

Test case 3 :

Input:
908172
Output:
27

Test case 4 :

Input:
-10001
Output:

2

Test case 5 :

Input:
999999
Output:
54

Test case 6 :

Input:
10000
Output:
1

Q9. Python Coding

Write a program to enter cost price and selling price of a product and find profit or loss incurred. If cost price and selling price are same then print *No Profit No Loss*.

Sample Input 1

7//Cost Price
9//Selling Price

Sample Output1

Profit=2

Sample Input 2

9
9

Sample Output 2

No Profit No Loss

Sample Input 3

100
90

Sample Output 3

Loss=10

ANSWER:Test case 1 :

Input:
7
9
Output:
Profit=2

Test case 2 :

Input:
9
9
Output:
No Profit No Loss

Test case 3 :

Input:
100
90
Output:
Loss=10

Test case 4 :

Input:
5
5
Output:
No Profit No Loss

Test case 5 :

Input:
6
4
Output:
Loss=2

Test case 6 :

Input:
100
120

Output:
Profit=20

Test case 7 :

Input:
600
200
Output:
Loss=400

Test case 8 :

Input:
400
400
Output:
No Profit No Loss